Loan Approval Analysis

Exploratory Data Analysis Report

Executive Summary

This report analyses 614 past loan applications to find out what drives loan approval and where a lender can approve more good loans while protecting itself from risky ones. About 69% of applications were approved. Credit history is by far the strongest factor: applicants whose credit history meets guidelines were approved 80% of the time, against 8% for those who did not meet guidelines, while income and loan size made almost no difference. 97 applicants who met the credit guidelines were still rejected, which represents about 14,900 (thousand) in requested loans and is the clearest opportunity to recover business. Semiurban applicants are the lender's strongest segment, and rural applicants are the biggest growth gap.

Objective

To understand which applicant characteristics drive loan approval, to find where the lender is losing profitable business or taking on avoidable risk, and to recommend practical actions that increase approvals of good loans and reduce risk.

Data Description

Dataset	Applications	Notes
train.csv	614	Includes the decision (Loan_Status: Y for approved, N for rejected)
test.csv	367	No decision column, used only so both files were cleaned with the same rules

All approval analysis uses the training file, because it is the only one with the outcome. The data follows the public Loan Prediction practice dataset: loan amounts are in thousands, loan terms are in months, and the credit history field records whether an applicant's credit history meets guidelines (1 for meets, 0 for does not meet). Income has no stated period or currency, so it was treated as monthly income. The analysis was done in Python using pandas, matplotlib and seaborn.

Data Cleaning and Preprocessing

Missing values in the training data:

Column	Missing values
Credit_History	50
Self_Employed	32
LoanAmount	22

Column	Missing values
Dependents	15
Loan_Amount_Term	14
Gender	13
Married	3

- **Categorical gaps** (gender, married, dependents, self-employed) were labelled "Unknown" instead of being filled with the most common value, which would invent data and inflate the largest group. Unknown groups were left out of the charts.
- **Loan amount** gaps were filled with the median (128 thousand), which is not distorted by very large loans, and **loan term** gaps were filled with the most common term (360 months).
- **Credit history** gaps were not guessed. They were kept as their own group ("Missing"), because we do not know whether those applicants met the guidelines.
- **Consistency:** the training and test files were combined, cleaned with the same rules and then separated again, using a source tag on each row.
- **Duplicates:** no duplicate loan IDs were found, and no applicant appears in both files.
- **New columns:** total household income (applicant plus co-applicant income), an estimated monthly repayment (loan amount divided by the number of months, with no interest included, so it is not a true EMI), and a repayment burden (estimated monthly repayment as a share of household income).

Key Findings

Overall approval

422 of 614 applications were approved, an approval rate of 68.7%.

Credit history is the gatekeeper

Credit history	Applicants	Approval rate
Meets guidelines	475	79.6%
Does not meet guidelines	89	7.9%

About 80% of all approved loans went to applicants whose credit history meets guidelines, and only 7 of the 89 applicants who did not meet guidelines were approved.

Location

Property area	Applicants	Approval rate	Share of approved loan value
Semiurban	233	76.8%	42.0%
Urban	202	65.8%	31.2%
Rural	179	61.5%	26.8%

Household factors

Characteristic	Approval rate
Married / not married	71.6% / 62.9%
Graduate / not graduate	70.8% / 61.2%
With a co-applicant / applicant only	71.8% / 64.8%
Self-employed / not self-employed	68.3% / 68.6%
Dependents: 0, 1, 2, 3 or more	69.0%, 64.7%, 75.2%, 64.7%

Self-employment makes almost no difference, and the number of dependents shows no clear pattern.

Income, loan size and affordability

- Income, loan size and repayment burden have almost no relationship with approval (correlations between -0.06 and 0.00).
- Approval rates by household income quarter range only from 67.5% to 71.2%.
- Approval by repayment burden does not fall as burden rises: the quarter with the heaviest burden was still approved 65.6% of the time, and the middle two quarters were approved most often (71.2% and 73.9%).

Where good loans are being rejected

97 applicants whose credit history meets guidelines were rejected. They requested about 14,900 (thousand) in total, roughly 154 thousand each. Among applicants who meet the guidelines, approval is lower for:

Segment	Approval rate (credit history meets guidelines)
Rural / Urban / Semiurban	71.5% / 76.8% / 87.7%
Not married / married	73.1% / 83.0%
Not graduate / graduate	74.7% / 80.8%

Loan term

85.7% of applications (526) are for the standard 360-month term, so demand is highly concentrated. The 180-month term has 44 applications (66% approved), and the 480-month term has only 15 applications with a 40% approval rate, which is too few to draw firm conclusions.

Insights

1. **Credit history decides the outcome.** Approval is 80% when credit history meets guidelines and 8% when it does not, while income and loan size barely matter.

2. **Semiurban is the strongest market.** It has the highest approval rate (77%) and 42% of approved loan value. Rural applicants are approved least often (61.5%) and make up only 27% of approved value, so rural is the biggest growth gap.
3. **Profitable loans are being rejected.** 97 applicants who meet the credit guidelines (about 14,900 thousand in requested loans) were turned away, more often when they were rural, unmarried or non-graduates.
4. **Stable households are favoured.** Married applicants, graduates and applications with a co-applicant are approved more often.
5. **Affordability is not a strong filter.** Applicants with the heaviest repayment burden are still approved about 66% of the time, so stretched borrowers may be getting through.
6. **Applicants with a missing credit history are approved at 74%,** nearly as often as those who meet the guidelines, even though their risk is unknown.
7. **The product is concentrated.** About 86% of applications are for one standard term.

Recommendations

1. **Fast-track applicants who meet the credit guidelines.** Build a quick approval path for applicants with a qualifying credit history and a healthy repayment burden, which speeds up decisions, lowers processing cost and wins customers before competitors do.
2. **Add a second-look review for rejected applicants who meet the guidelines,** especially rural, unmarried and non-graduate applicants. Recovering even a share of the 97 rejected loans is direct revenue gain.
3. **Grow semiurban lending and build a rural strategy.** Put marketing weight behind semiurban areas, where approval and loan value are highest, and design rural products (for example community guarantors or terms that follow seasonal income) to raise the 61.5% rural approval rate safely.
4. **Encourage joint applications.** Offer a co-applicant option to borderline applicants, because applications with a co-applicant are approved more often and add repayment capacity.
5. **Add an affordability check next to the credit check.** Set a cap on the repayment burden (estimated monthly repayment as a share of income) so stretched borrowers are screened out and defaults are reduced.
6. **Set a clear policy for applicants with a missing credit history.** Approving them at nearly the same rate as applicants who meet the guidelines carries hidden risk. Use smaller starter loans or extra documentation, and treat them as a customer acquisition opportunity.
7. **Test other loan terms.** With 86% of demand on one term, pilot shorter or more flexible terms to reach applicants the standard product does not serve.

Limitations

- The sample is small (614 applications), so smaller groups, such as the less common loan terms, should be read with caution.
- The findings show associations and not causes. For example, credit guidelines is strongly linked to approval, but the data does not show how the lender weighed each factor.
- 8.1% of applicants have a missing credit history, which limits what can be said about that group.

Conclusion

Loan approval in this portfolio is driven overwhelmingly by credit guidelines, with household stability (marriage, education, a co-applicant) and location playing a supporting role, while income and loan size matter very little. The business opportunity has two parts: recover profitable loans that are rejected despite a qualifying credit history, and tighten risk control where approvals ignore affordability or rely on missing credit data. Acting on these recommendations would grow the loan book in its strongest segments, open up the rural market and reduce exposure to risky borrowers. A sensible next step is to build a predictive model on this data to automate and test these decision rules.